

PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

**ENTERPRISE WATER DISTRIBUTION AND FLEET MANAGEMENT
PLATFORM FOR OPTIMIZED OPERATIONS AND COST EFFICIENCY**

In partial fulfillment for the requirements in capstone 1

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Water is a fundamental resource that supports human survival, economic activity, and environmental sustainability. It is essential in households, businesses, agriculture, and industrial operations. As populations grow and urban areas expand, the demand for reliable water services continues to increase, requiring more efficient systems for distribution and management. According to the United Nations, sustainable water management is a critical component of urban development, especially in rapidly growing cities where demand often exceeds system capacity.

Water distribution involves the organized delivery of clean water from supply sources to end users through infrastructure such as pipelines, pumping systems, storage facilities, and operational coordination. While infrastructure plays a major role, operational efficiency is equally important in ensuring that water is delivered consistently and without delays. Inefficient distribution systems may lead to service interruptions, resource wastage, and increased operational costs, especially in expanding urban communities.

In the Philippines, increasing urbanization has placed significant pressure on utility service providers. Cities continue to expand due to population growth and economic development, resulting in higher demand for essential services such as water supply. In particular, Iloilo City has experienced continuous growth in residential subdivisions, commercial establishments, and industrial activities. This expansion has increased water consumption and created operational challenges for water distribution providers that must meet rising service expectations.

Despite improvements in infrastructure, many water delivery operations still rely on traditional management practices. Scheduling deliveries is often done through manual coordination such as phone calls, handwritten logs, and spreadsheet-based tracking. While these methods may be manageable for small-scale operations, they become inefficient as the number of customers increases. Manual systems may result in delayed updates, scheduling conflicts, and difficulty maintaining accurate operational records.

Another operational issue is inefficient coordination of delivery routes. In urban areas like Iloilo City, traffic congestion, road construction, and increasing vehicle density contribute to delays in delivery operations. Without optimized routing, delivery vehicles may take longer travel times, consume more fuel, and experience inconsistent scheduling. These inefficiencies increase operational costs and reduce service reliability.

Fleet monitoring is also a major challenge in traditional systems. Many organizations lack real-time visibility of delivery vehicles and depend on manual updates from drivers. This limits the ability of managers to track delivery progress, respond to delays, and ensure efficient operations. In some cases, problems are only identified after delays or customer complaints occur. Studies emphasize that real-time monitoring systems improve logistics coordination and operational transparency (Cominola et al., 2019).

Operational costs are also affected by inefficiencies in scheduling and fleet management. Poor routing, delayed deliveries, and lack of coordination can lead to excessive fuel consumption, higher maintenance expenses, and wasted operational time. These issues reduce overall productivity and place additional financial pressure on service providers. As a result,

organizations are increasingly exploring technology-based solutions to improve efficiency and reduce costs.

Modern enterprise systems offer improved solutions through automation, centralized data management, and real-time monitoring. These systems allow organizations to manage delivery schedules, track fleet performance, and analyze operational data more effectively. According to Boulos and Perelman (2021), digital water management systems improve decision-making and operational efficiency by providing better visibility and control over distribution processes.

In response to these challenges, the proposed study titled *Enterprise Water Distribution and Fleet Management Platform for Optimized Operations and Cost Efficiency* aims to develop a centralized system that integrates water distribution and fleet management operations. The system is designed to automate scheduling, optimize delivery routes, and provide real-time monitoring of delivery activities. This helps reduce delays, improve coordination, and enhance operational performance.

Unlike traditional manual systems, the proposed platform aims to provide a more structured and technology-driven approach to managing water delivery operations. It is expected to improve communication between dispatchers and drivers, reduce operational inefficiencies, and support better decision-making through centralized data access and monitoring tools.

Overall, the integration of water distribution and fleet management into a single platform is expected to improve service reliability and operational efficiency in growing urban areas such as Iloilo City. The study seeks to address existing limitations in manual systems and provide a more scalable and efficient solution for modern water distribution challenges.

1.2 Statement of the Problem

While water distribution is essential for urbanized areas like Iloilo City, the current logistical and operational frameworks face significant unresolved challenges. This study seeks to address the following problems:

Specifically, it seeks to answer the following questions:

1. How are current water distribution and fleet operations managed in terms of delivery scheduling, trip recording, fuel usage documentation, and delivery reporting?
2. What problems are encountered in the current system in terms of manual data recording, difficulty in tracking fuel consumption, lack of structured trip documentation, delays in consolidating operational data, and inefficiencies in monitoring delivery performance?
3. How does the absence of a centralized system affect the accuracy of operational records, communication between drivers and administrators, monitoring of fleet performance, and decision-making in water distribution operations?
4. What system features can be developed to address these issues, specifically a driver-based input system for trip data encoding, centralized data management for administrators, and automated reporting and analytics for operational performance?
5. How can the proposed system improve operational efficiency, enhance data accuracy, support cost monitoring (particularly fuel usage), and strengthen decision-making in water distribution and fleet management operations?

1.3 OBJECTIVE OF THE STUDY

1.3.1 GENERAL OBJECTIVE

The general objective of this study is to design and develop a centralized fleet management and water distribution system that utilizes **system monitoring** and automated routing to optimize logistical efficiency, enhance service reliability, and reduce operational costs.

1.3.2 Specific Objectives

To achieve this goal, the study aims to:

1. Assess the logistical requirements and existing limitations of manual water distribution methods in Ilo-Ilo City areas.
2. Design a secure and intuitive user interface for both fleet managers and delivery personnel to monitor active operations.
3. Develop an automated routing algorithm to optimize delivery paths and minimize fuel consumption.
4. Implement system-based monitoring and resource tracking features to provide visibility into fleet operations, delivery progress, and cost-related data.to provide full visibility into fleet locations and delivery status.
5. Construct a centralized administrative module for the management of vehicle schedules, driver assignments, and demand forecasting.
6. Manage operational data, including delivery logs and fleet maintenance records, using a secure and structured database.
7. Evaluate the system's effectiveness in reducing delivery delays, improving resource utilization, and maintaining performance under peak demand.

1.4 Scope and Delimitation

The primary focus of this study is the development of a Web System designed for the centralized management of fleet operations and water distribution logistics. The system is specifically engineered for internal use by authorized personnel and the business owner to streamline high-traffic distribution corridors.The system focuses on **cost monitoring, operational**

tracking, and route optimization using system-encoded data and trip logs. It does not utilize GPS-based real-time tracking; instead, monitoring is based on recorded inputs, delivery reports, and administrative updates within the system.

1.5 Significance of the Study

The proposed system will benefit the following:

- **Water Distribution Owners** – by providing a high-level overview of operations through analytics and reports, allowing for better cost control and informed decision-making.
- **Authorized Personnel (Administrators)** – by streamlining the complexity of scheduling and driver management, reducing manual errors, and enabling effective oversight of the delivery process.
- **Fleet Drivers** – by providing organized schedules and optimized routes that reduce travel time, fuel consumption, and the stress associated with uncoordinated logistics.
- **Iloilo City Community** – by indirectly improving the reliability, consistency, and timeliness of water delivery services through more efficient resource allocation in rapidly urbanizing city zones.

1.6 Definition of Terms

- **Automated Routing Algorithm** – a computational procedure used to determine the most efficient path for delivery vehicles based on distance, traffic conditions, and delivery priorities.
- **Centralized Administrative Module** – a primary control interface within the Web application where the owner and authorized personnel manage all system data, including driver profiles and fleet status.
- **Database Management System (DBMS)** – the software used to store, retrieve, and manage operational data such as delivery logs, vehicle maintenance records, and user credentials.
- **Fleet Management** – the administrative approach used to coordinate a company's vehicles to ensure they are used efficiently, safely, and maintained properly.

- **Logistical Efficiency** – the measurement of how effectively resources, such as fuel and time, are utilized during the distribution of water to minimize waste and cost.
 - **Operational Overheads** – the ongoing expenses related to the daily running of the business, specifically focusing on fuel costs, vehicle repairs, and labor hours in this study.
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CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

Miller and Thompson (2021) discussed the application of automated routing algorithms in urban logistics platforms. Their study revealed that optimization models improved delivery accuracy by 25% by analyzing traffic patterns and historical fuel consumption data. Similarly, Chen et al. (2020) emphasized that fleet monitoring systems significantly enhance operational transparency and stakeholder trust in resource distribution applications. Furthermore, Wang (2022) noted that centralized Web-based management systems are preferred for administrative oversight in small-to-medium enterprises due to their robust data handling capabilities.

2.2 Local Studies

According to **Dela Cruz and Santos (2022)**, logistics-based systems developed in Philippine academic settings improve process automation and data accuracy, particularly in the context of local delivery services. Their study on intelligent monitoring applications highlights the importance of localized data in overcoming unique regional infrastructure challenges. **Reyes (2021)** also emphasized that integrating data-driven decision-making into water utility

management enhances system reliability and reduces the response time to sudden spikes in community demand, which is critical in rapidly urbanizing Philippine cities like Iloilo.

2.3 Related Literature

Logistics management literature explains that route optimization relies on advanced algorithmic modeling to determine the most efficient and cost-effective delivery paths (Toth & Vigo, 2021). In the realm of software engineering, Pressman (2019) emphasizes the necessity of structured development methodologies to guarantee system reliability, particularly when integrating real-time data analytics and cloud-based tracking APIs. Furthermore, research onunderscores the critical role of operational visibility and fuel consumption monitoring in minimizing overhead costs and preventing the underutilization of corporate assets (Goldsby et al., 2018).

2.4 Synthesis of the Reviewed Literature

The reviewed foreign and local studies demonstrate that automated routing and real-time monitoring significantly improve logistical intelligence and operational decision-making. However, much of the existing research focuses on large-scale commercial logistics or consumer-facing delivery apps, with limited focus on specialized, internal Web systems designed for water distribution management in rapidly urbanizing local contexts like Iloilo City. This study addresses this gap by integrating monitoring-based tracking and cost analysis and automated scheduling into a centralized administrative platform. By providing the owner and authorized personnel with direct oversight of distribution costs and fleet efficiency, the study aligns with current technological trends while providing a practical solution to localized infrastructure challenges.

CHAPTER 3

METHODOLOGY

3.1 Research Design

The study will use a **developmental research design**, focusing on the design, development, and evaluation of the proposed **Enterprise Water Distribution and Fleet Management Platform**. This design follows a systematic process to create a functional software solution that addresses the logistical inefficiencies identified in Iloilo City.

3.2 System Development Methodology

The Waterfall Model will be used, consisting of the following phases:

1. Requirements Analysis
 2. System Design
 3. Implementation
 4. Testing
 5. Deployment (Prototype)
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3.3 System Architecture

A dedicated, responsive portal where users perform administrative tasks, monitor fleet movements in real-time, and manage cost oversight. By moving to a web-based model, the system ensures seamless access across the internal network without the need for local software installations.

3.4 Tools and Technologies Used

- **Frontend:** Html,css and JavaScript
 - **Backend:.**Python flask
 - **Database:** Firebase
 - **Development Tools:** Visual Studio Code
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3.5 Data Gathering Techniques

- Interview with the Client
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3.6 Testing Procedures

- User Acceptance Testing (UAT)
 - Unit Testing
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3.7 Evaluation Criteria:

The system will be evaluated based on the following criteria:

- Functionality
- Usability
- Security

SUPPORTING DIAGRAMS (Included within Chapter 3)

Data Flow Diagram (DFD – Level 0)

- User (Admin) → Water Distribution System → Database
 - User submits data
 - System processes data
 - Results are stored and retrieved
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3.11 Use Case Diagram

Actors:

- Administrator (Main Client)
- System (Machine Learning Engine)
- Fleet Staff (Drivers)

Use Cases:

- Manage Fleet Vehicles (Admin)
 - Input Water Orders (Admin)
 - Generate ML Cost Predictions (System)
 - Optimize Delivery Routes (System)
 - Monitor Fuel Expenses (Admin)
 - Manage User Accounts (Admin)
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End of Proposal

Note to the instructors handling this subject

CHED BSIT ALIGNMENT STATEMENT

This capstone project complies with CHED BSIT standards by demonstrating competencies in application development, database management, systems analysis and design, and emerging technologies such as machine learning. The project follows outcomes-based education principles and integrates ethical considerations, user safety, and data privacy.

